Introduction to Telecommunications
EN1052

Chathuranga Weeraddana

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Some Key Properties of FT

Recall the Essence of Systems

- we use the following notation:
- $x(t)$: time domain signal
- $X(f)$: FT of $x(t)$
- $x(t)$ $X(f)$ are called FT pairs

$G(f)$ is Complex

- let $g(t)$ $G(f)$ be FT pairs
- since $G(f)$ is complex, we have

$$G(f) = |G(f)|e^{j\theta_g(f)} \tag{1}$$

- $|G(f)|$: the amplitude spectra
- $\theta_g(f)$: the phase spectra
- determine amplitude and phase spectra of previous example

Conjugate Symmetry Property

- if $g(t)$ is a real signal , then we have

$$G(-f) = G^*(f) \tag{2}$$

- therefore, we have
  - $|G(-f)| = G(f)$, i.e., amplitude spectra is an even function
  - $\theta_g(-f) = -\theta_g(f)$, i.e., phase spectra is an odd function

Transform of Some Useful Functions

- reading assignment: § 3.2 of [Lathi]
- a short table of FT pairs are listed in Table 3.1 of [Lathi]

Some Properties of the FT

- read: § 3.3 of [Lathi]
- a short table of FT pairs are listed in Table 3.2 of [Lathi]

Some Properties of the FT

- example: time-scaling property
- suppose $\mathcal{F}[g(t)] = G(f)$
- then we have

$$\mathcal{F}[g(at)] = \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

Some Properties of the FT

- significance of time-scaling
  - time compression (i.e., $|a| > 1$) results in its spectral expansion
  - time expansion (i.e., $|a| < 1$) results in its spectral compression
  - compression means rapid variation of signal
  - synthesizing such a signal require sinusoidal components of larger frequencies
  - e.g., compare $g(t) = \cos(4\pi f_0 t)$ and $h(t) = \cos(2\pi f_0 t)$

Some Properties of the FT

- example: time-shifting property
- suppose $\mathcal{F}[g(t)] = G(f)$
- then we have

$$\mathcal{F}[g(t - t_0)] = G(f)\, e^{-j2\pi f t_0}$$

Some Properties of the FT

- significance of time-shifting
  - delaying a signal by t_0 does not change its amplitude spectrum
  - however, the phase spectrum is changed by $-2\pi f t_0$
  - see Figure. 3.18 [Lathi]

Some Properties of the FT

- example: frequency-shifting property
- suppose $\mathcal{F}[g(t)] = G(f)$
- then we have

$$\mathcal{F}\left[g(t)e^{j2\pi f_0 t}\right] = G\left(f - f_0\right)$$

Some Properties of the FT

- significance of frequency-shifting
  - frequency shifting in practice is achieved by $g(t)\cos(2\pi f_0 t)$
  - discuss!.. known as amplitude modulation
  - different radio stations use different f_0 values
- $g(t)$ is called the modulating signal
- $\cos(2\pi f_0 t)$ is called the carrier and f_0 is the carrier frequency
- $g(t)\cos(2\pi f_0 t)$ is called the modulated signal

Some Properties of the FT

- example: convolution Theorem
- recall $g(t) * w(t) = \int_{-\infty}^{\infty} g(\tau) w(t-\tau) d\tau$
- suppose $\mathcal{F}[g(t)] = G(f)$ and $\mathcal{F}[w(t)] = W(f)$
- we have the time convolution as

$$\mathcal{F}\left[g(t) * w(t)\right] = G\left(f\right) W\left(f\right)$$

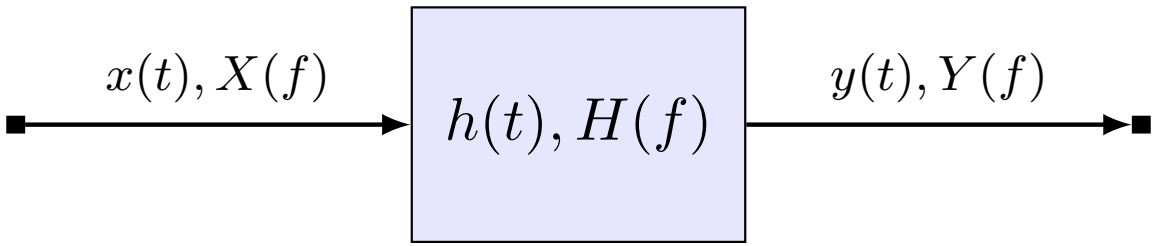
- we have the frequency convolution as

$$\mathcal{F}\left[g(t) w(t)\right] = G\left(f\right) * W\left(f\right)$$

Some Properties of the FT

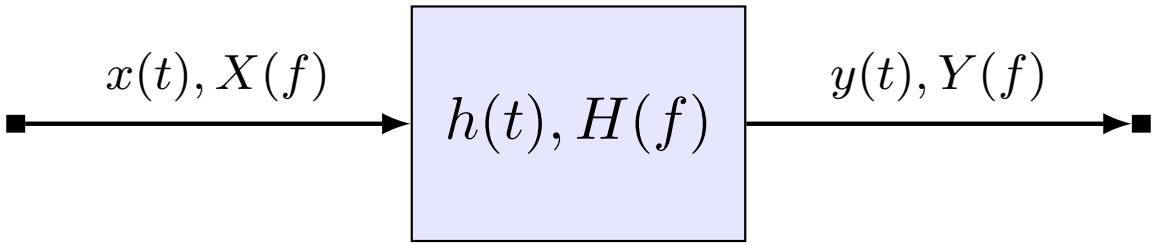
- significance of time convolution
  - spectrum of the output signal of an LTI system
- significance of frequency convolution
  - bandwidth of the product of two signals
  - see p. 88 of [Lathi]

Signal Through a LTI System



- $y(t) = h(t) * x(t)$
- $Y(f) = H(f)X(f)$, $H(f)$ is usually referred to as transfer function or frequency response of the LTI system

Signal Distortion During the Transmission



- we have

$$Y(f) = H(f)X(f) = |H(f)||X(f)|e^{j\left(\theta_h(f)+\theta_x(f)\right)}$$

- thus,
  - $|X(f)|$ has changed to $|H(f)||X(f)|$
  - $\theta_x(f)$ has changed to $\theta_h(f) + \theta_x(f)$